EXP NO: 9 EXPERIMENT TO UNDERSTAND LOGISTIC REGRESSION FOR A GIVEN DATA SET

AIM:

Understand the Logistic Regression algorithm for the dataset given.

PYTHON CODE:

```
import numpy as np
import pandas as pd

df = pd.read_csv('Social_Network_Ads.csv')
df.head()
features = df.iloc[:, [2, 3]].values
label = df.iloc[:, 4].values
features
label
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
for i in range(1, 401):
    x_train, x_test, y_train, y_test = train_test_split(
        features, label, test_size=0.2, random_state=i
    )

    model = LogisticRegression()
    model.fit(x_train, y_train)
    train_score = model.score(x_train, y_train)
    test_score = model.score(x_test, y_test)
    if test_score > train_score:
        print("Test {} Train {} Random State {}".format(
            test_score, train_score, i
        ))
x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=4
)
finalModel = LogisticRegression()
finalModel.fit(x_train, y_train)
print(finalModel.score(x_train, y_train))
print(finalModel.score(x_test, y_test))
from sklearn.metrics import classification_report
print(classification_report(label, finalModel.predict(features)))
```

OUTPUT:

```
Test 0.8625 Train 0.8375 Random State 294
Test 0.8875 Train 0.828125 Random State 297
Test 0.8625 Train 0.834375 Random State 300
Test 0.8625 Train 0.85 Random State 301
Test 0.8875 Train 0.85 Random State 302
Test 0.875 Train 0.846875 Random State 303
Test 0.8625 Train 0.834375 Random State 305
Test 0.9125 Train 0.8375 Random State 306
Test 0.875 Train 0.846875 Random State 308
Test 0.9 Train 0.84375 Random State 311
Test 0.8625 Train 0.834375 Random State 313
Test 0.9125 Train 0.834375 Random State 314
Test 0.875 Train 0.8375 Random State 315
Test 0.9 Train 0.846875 Random State 317
Test 0.9125 Train 0.821875 Random State 319
Test 0.8625 Train 0.85 Random State 321
Test 0.9125 Train 0.828125 Random State 322
Test 0.85 Train 0.846875 Random State 328
Test 0.85 Train 0.8375 Random State 332
Test 0.8875 Train 0.853125 Random State 336
Test 0.85 Train 0.8375 Random State 337
Test 0.875 Train 0.840625 Random State 343
Test 0.8625 Train 0.84375 Random State 346
Test 0.8875 Train 0.83125 Random State 351
Test 0.8625 Train 0.85 Random State 352
Test 0.95 Train 0.81875 Random State 354
Test 0.8625 Train 0.85 Random State 356
Test 0.9125 Train 0.840625 Random State 357
Test 0.8625 Train 0.8375 Random State 358
Test 0.85 Train 0.840625 Random State 362
Test 0.9 Train 0.84375 Random State 363
Test 0.8625 Train 0.853125 Random State 364
Test 0.9375 Train 0.821875 Random State 366
Test 0.9125 Train 0.840625 Random State 369
Test 0.8625 Train 0.853125 Random State 371
Test 0.925 Train 0.834375 Random State 376
Test 0.9125 Train 0.828125 Random State 377
Test 0.8875 Train 0.85 Random State 378
Test 0.8875 Train 0.85 Random State 379
Test 0.8625 Train 0.840625 Random State 382
Test 0.8625 Train 0.859375 Random State 386
Test 0.85 Train 0.8375 Random State 387
Test 0.875 Train 0.828125 Random State 388
Test 0.85 Train 0.84375 Random State 394
Test 0.8625 Train 0.8375 Random State 395
Test 0.9 Train 0.84375 Random State 397
Test 0.8625 Train 0.84375 Random State 400
0.840625
0.9
              precision    recall  f1-score   support

           0       0.87      0.91      0.89       257
           1       0.82      0.75      0.78       143

    accuracy                           0.85       400
   macro avg       0.84      0.83      0.84       400
weighted avg       0.85      0.85      0.85       400
```

RESULT:

Thus the code is executed successfully.